Phantom Problems in Science [2]

The world is teeming with problems. Wherever man looks, some new problem crops up to meet his eye—in his home life as well as in his business or professional activity, in the realm of economics as well as in the field of technology, in the arts as well as in science. And some problems are very stubborn; they just refuse to let us in peace. Our agonized thinking of them may sometimes reach such a pitch that our thoughts haunt us throughout the day, and even rob us of sleep at night. And if by lucky chance we succeed in solving a problem, we experience a sense of deliverance, and rejoice over the enrichment of our knowledge. But it is an entirely different story, and an experience annoying as can be, to find after a long time spent in toil and effort, that the problem which has been preying on one's mind is totally incapable of any solution at all—either because there exists no indisputable method to unravel it, or because considered in the cold light of reason, it turns out to be absolutely void of all meaning—in other words, it is a *phantom problem*, and all that mental work and effort was expended on a mere nothing. There are many such phantom problems—in my opinion, far more than one would ordinarily suspect—even in the realm of science.

There is no better safeguard against such unpleasant experiences than to ascertain in each instance, and at the very outset, whether the problem under consideration is a genuine or meaningful one, and whether a solution for it is to be expected. In view of this situation I will cite and examine a number of problems, in order to see whether they happen to be mere phantom problems. By doing so, I may be able to render a genuinely useful service to some of you. My selection of these problems to be exhibited as specimens is not based on any systematic viewpoint, and even less can it lay a claim to completeness in any respect. Most of them are taken from the realm of science, because this is the field in which the relevant factors are the most clearly discernible. However, this consideration will not deter me from touching upon other fields, too, whenever I can reasonably surmise that the subject holds an interest for you.

I.

In order to decide whether or not a given problem is truly meaningful, we must first of all examine closely the assumptions contained in its wording. In many instances, these alone will immediately reveal the problem under